Habib University
shaping futures

Khidmat Proposal Form

Section 1: Metadata

1.1. Project Information

Title: Clinical ICD Coding System for SIUT	
Start: 1st June	End: 31st August

1.2. Student(s) Information

Name: Hafsa Ehsan	ID:09925
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Name: Areesha Ashfaq	ID: 09879
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1.3. Client Information

Organization Name: Sindh Institute of Urology And Transplantation	Contact: 021-36242366
Address: Sardar Yaqoob Ali Khan Road, Near Civil Hospital, Nanak Wara, Karachi, 74200, Pakistan	
Supervisor: Ashar Mahmood Minai	Cell: 0300-8203797
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Section 2: The Project

2.1. Project Description:

SIUT processes a high volume of patient records daily where every discharge summary written by a doctor needs to be reviewed by a trained medical coder and converted into ICD-10 codes, these are standard structured codes that classify the principal diagnosis, any associative diagnoses, procedures performed and medications administered. ICD codes are a globally standardised system maintained by the WHO that assigns a unique alphanumeric code to every known disease, diagnosis, and medical procedure. This is currently done entirely by hand, which is slow and can produce inconsistencies across coders over time.

This project builds a coding support system for SIUT's coding team. The system reads a doctor's discharge summary and suggests the appropriate ICD codes for each category along with a confidence score and a reference to the exact text in the note that supports each suggestion. The coder reviews every suggestion and makes the final decision; the system does not assign codes on its own.

The core of the project is building a complete database system from scratch. There is no existing database at SIUT for this workflow. The team will design and implement the full database schema, import the WHO ICD10 reference data and build all the logic that populates and queries the database as the system runs. This covers storing incoming records, saving generated suggestions with their evidence, logging every coder decision as a permanent trail and feeding corrected data back into the model for periodic retraining.

On top of the database, the project will include:

- A text processing pipeline that cleans and parses discharge summaries, extracts medical concepts, maps them to standard ICD codes and classifies each finding as a principal or associative diagnosis.
- A backend API that connects the processing pipeline to the database and to the user interfaces.
- A coder interface where staff can view incoming records, review suggestions side by side with the original note, accept or edit codes and ask follow up questions about any suggestion through a built in chat panel.
- An admin dashboard where they can monitor how many records have been processed, how often suggestions are being accepted vs corrected and view a full audit log.

The system will be built entirely using open-source tools, all free of charge. The table below lists what is being used and what each component does.

Steps	Tools	Purpose
Database	PostgreSQL	Stores all records, suggestions, coder decisions and audit logs. The WHO ICD-10 reference is also imported here as a local lookup table.
Backend	Python, FastAPI	Connects the processing pipeline to the database and the user interfaces through a REST API.
Text Processing	Python, MedspaCy, RapidFuzz	Reads discharge summaries, extracts medical concepts, corrects spelling errors and maps findings to ICD codes.
ICD Reference	WHO ICD-10 (Excel Sheet)	The official code list, imported once into the database and used for all code lookups.
User Interface	Python, PyQt6, Matplotlib	Desktop application for coders and the admin dashboard with charts.

2.2. Expected Deliverables:

The following will be handed over to SIUT at the end of the project:

Database System

- A fully designed database built from scratch, covering all tables needed for records, suggestions, coder decisions and audit logs.
- The complete WHO ICD10 reference imported into the database as a queryable local table.
- An automated retraining pipeline that takes coder corrections from the database, retrains the model, validates it and deploys the improved version without manual intervention.

Text Processing & Code Assigning

- A text cleaning and section parsing module that handles the inconsistent formatting found in real SIUT discharge summaries, including typos, missing punctuation and duplicate entries.
- A medical entity extraction module using a pre-trained clinical NLP library.
- An ICD code lookup and classification module that queries the local database and returns the top candidate codes with confidence scores and evidence references for each finding.
- A medication extraction module that parses the medication lists in discharge summaries into structured drug and route entries.

User Interfaces

- A coder desktop application, which will be showing the required outcomes like record queue, a coding workspace showing the suggestions alongside diagnosis and a coder history view.
- A built in chat panel in the coding workspace where coders can ask questions like “why did you suggest this as the principal diagnosis?” and receive answers pointing back to the original note.
- An admin dashboard with suggestion acceptance rates, most corrected ICD categories and a searchable audit log for the coding done.

2.3. Planned Schedule: .

Starting Date: 1st June 2026
Ending Date: 31st August 2026

Weeks 1–2	**Documentation & Setting up**	Reviewing & understanding sample records and coding manuals. Set up the development environment on SIUT approved infrastructure and fulfill any official prior requirements.
Weeks 3–5	**Database & Pipeline**	Design and build PostgreSQL schema, import WHO ICD-10 reference data, set up database. Build text cleaning, section parsing, and medical entity extraction pipeline. Build ICD code lookup, confidence scoring, principal vs associative classification, medication extraction.
Weeks 6–8	**Interfaces**	Build the coder desktop application and admin dashboard. Implement the chat panel with database backed question answering. Connect all interfaces to the backend API.
Weeks 9–10	**Trial & Refinement**	SIUT coders use the system alongside their existing workflow on a controlled set of records. Weekly feedback sessions. Corrections feed into the retraining loop. Accuracy tracked across retraining cycles.
Weeks 11–12	**Handover**	Final fixes and tuning. Complete all documentation and the final project report. Training session for SIUT IT and coding staff. Full source code and model transfer.

2.4. Modus Operandi: *Kindly explain the mode of operation i.e., how often and for how long should the student should visit your site, can they work remotely, how you will supervise them*

The work will done in a hybrid arrangement. On site visits to SIUT when necessary and day to day development work will be done remotely. Any work involving patient data will be carried out exclusively on infrastructure provided or approved by SIUT.

A weekly review meeting with the external supervisor will be held throughout the project to track progress. A brief written progress update will be shared each week covering what was completed, what is in progress and any open questions.

2.5. Comments: *Kindly mention any special requirements you expect and/or any comments you might have.*

Name and signature of student	Location and date
Ashar Mahmood Minai	SIUT, Karachi, 15-May-2026
Name and signature of external supervisor	Location and date
Name and signature of internal supervisor	Location and date